

llmXive Automated Review of Agentic Environment Engineering for Large Language Models: A Survey of Environment Modeling, Synthesis, Evaluation, and Application

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

This survey contains multiple fatal factual inaccuracies regarding the existence of the models and benchmarks it cites as foundational evidence. The paper repeatedly references specific model versions (e.g., GPT-5.4, Gemini-3.1-Pro, Kimi K2.5, DeepSeek-V3.2) and benchmarks (e.g., AI_Idea_Bench_2025) with publication dates in 2025 and 2026. As of the current date, these models and papers do not exist in the public record.

The introduction explicitly states that “Representative advanced models such as GPT-5.4... Gemini-3.1-Pro... and Kimi K2.5 have demonstrated strong agentic capabilities.” This is a direct factual claim that is unsupported because the cited entities are hallucinated. Similarly, the bibliography contains entries for papers dated in 2026 (e.g., LongCat-Flash-Thinking-2601, Kimi_K2.5) with arXiv IDs that follow a future numbering scheme (e.g., abs/2602.02276).

In a survey paper, the validity of the taxonomy and the discussion of “state-of-the-art” capabilities relies entirely on the existence of the cited works. Citing non-existent future models as evidence for current capabilities breaks the chain of evidence completely. The paper cannot be trusted to accurately reflect the current landscape of agentic environments if its primary examples are fabricated. This is not a matter of overstatement but of citing non-existent sources to support load-bearing claims about the field’s current state.

Action items.

- **[fatal]** This survey contains multiple fatal factual inaccuracies regarding the existence of the models and benchmarks it cites as foundational evidence. The paper repeatedly references specific model versions (e.g., GPT-5.4, Gemini-3.1-Pro, Kimi K2.5, DeepSeek-V3.2) and benchmarks (e.g., AI_Idea_Bench_2025) with publication dates in 2025 and 2026. As of the current date, these models and papers do not exist in the public record. The introduction explicitly states that “Representative advanced models suc

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 provides a comprehensive visual taxonomy of agentic environments but suffers from legend inconsistencies where defined categories like ‘Agent Evolution’ are missing from the plot, and color coding for ‘Env Synthesis’ methods is ambiguous.

Figure 2

Figure 2 effectively visualizes the comparison between Data Engineering and Environment Engineering using clear icons and directional arrows. The layout is uncluttered, and the caption accurately describes the content shown.

Figure 3

Figure 3 effectively visualizes eight distinct pairs of environment attributes using clear icons and text labels. The layout is organized, the visual metaphors are intuitive, and the caption accurately describes the content without requiring external definitions.

Figure 4

Figure 4 provides a clear, well-organized visual overview of the seven environment domains listed in the caption. The use of distinct colors, icons, and labeled sub-sections effectively communicates the scope of each domain without clutter or ambiguity.

Figure 5

Figure 5 effectively visualizes the three symbolic environment synthesis methods described in the caption. The diagram is clear, with distinct color-coded sections for each method and legible labels for all components and processes.

Figure 6

Figure 6 effectively visualizes the three neural environment synthesis paradigms (Pixel-Level, Word-Level, and Latent-Level) described in the caption. The diagram is clear, with distinct sections for each modeling approach and intuitive icons representing inputs, processing, and outputs.

Figure 7

Figure 7 is a clear and well-organized conceptual diagram that effectively visualizes the four agent evolution paradigms described in the caption. The use of distinct color-coded quadrants and consistent iconography makes the complex taxonomy easy to navigate and understand.

Figure 8

Figure 8 is a clear and well-organized schematic that effectively visualizes the three categories of environment evolution paradigms described in the caption. The layout is logical, the icons are distinct, and the text labels are legible, making the figure easy to understand without ambiguity.

Action items.

- **[science]** Figure 1: The legend defines ‘Env Symbolic Syn’ (pink) and ‘Env Neural Syn’ (purple), but the timeline groups labeled ‘Env Synthesis’ (top center) contain boxes of both colors, creating ambiguity about which specific methods are symbolic versus neural.
- **[writing]** Figure 1: The timeline axis labels (2023-2026) are positioned on the left, but the data points (years) are scattered across the entire width; the visual connection between the specific year markers on the axis and the corresponding clusters of boxes is unclear.
- **[writing]** Figure 1: The legend includes ‘Agent Evolution’ (light green), but there are no corresponding light green boxes in the main chart area, suggesting a mismatch between the legend and the plotted data.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper is generally well-structured for a survey, but it relies on several acronyms and symbols that are either undefined or defined too late for a competent reader from an adjacent field (e.g., a researcher in NLP or Robotics who is not deeply embedded in the specific sub-ecosystems of “Model Context Protocol” or “Joint Embedding Predictive Architecture”).

The most significant barrier is the use of **MCP** (Model Context Protocol) throughout the text (Introduction, Section 3, Section 4, Section 5, Section 6) without ever spelling it out. While this is becoming a standard term in the specific “agentic tool” niche, it is not yet general knowledge across the broader LLM field. A reader from a neighboring discipline would likely stall here, unable to parse the distinction between “MCP-based” and other tool-use paradigms.

Additionally, the paper uses **PDDL** (Planning Domain Definition Language) in the context of symbolic synthesis. While standard in classical planning, it is not always assumed knowledge in modern LLM agent surveys. A one-sentence gloss would suffice.

The use of custom LaTeX macros (`\Text`, `\Image`, `\Yes`, `\No`) in the tables (Section 4) is a technical jargon issue for the reader of the source or a raw text extraction. These symbols are not standard LaTeX and are not defined in the preamble provided in the snippet, rendering the table content ambiguous without the compiled PDF.

Finally, **JEPA** (Joint Embedding Predictive Architecture) is used frequently in the Neural Synthesis section. While a specific architecture, it is not as foundational as “Transformer” or “CNN”. Defining it at first use would prevent confusion with other “J” or “P” acronyms.

These are all low-cost fixes (adding a parenthetical expansion) that significantly improve accessibility for the target “adjacent-field PhD” audience.

Action items.

- **[writing]** Section 1 (Introduction) and Section 3 (Environment Attribute) use the acronym ‘MCP’ (e.g., ‘OSWorld-MCP’, ‘MCPVerse’, ‘MCP-based Tool Use’) without ever defining it. An adjacent-field reader cannot infer that this refers to the ‘Model Context Protocol’. Expand at first use: ‘Model Context Protocol (MCP)’.
- **[writing]** Section 3.2 (Symbolic vs. Neural) and Section 5.1 (Symbolic Synthesis) reference ‘PDDL’ (e.g., ‘PDDL, AgentWorldModel’) without definition. While standard in planning, it is not universal in LLM agent literature. Add a brief gloss: ‘Planning Domain Definition Language (PDDL)’.
- **[writing]** Section 4.1 (GUI) and Table 1 use custom LaTeX macros like ‘\Text’, ‘\Image’, ‘\Video’, ‘\Yes’, and ‘\No’ in the table body. These are not standard LaTeX commands and are undefined in the provided source, making the table content opaque to a reader compiling or reading the raw text. Define these macros or replace with standard text (e.g., ‘Text’, ‘Image’).
- **[writing]** Section 6.3 (Training Reward Design) introduces the acronym ‘RPRM’ (‘robot process reward model (RPRM)’) but uses it later without re-expansion or ensuring the definition is prominent. Ensure ‘RPRM’ is clearly defined at first use and consider if the acronym is necessary given the term is used only a few times.
- **[writing]** Section 5.2 (Neural Synthesis) and Section 5.3 (Latent-level Modeling) use ‘JEPA’ (e.g., ‘I-JEPA’, ‘seq-JEPA’, ‘V-JEPA 2’) without defining the acronym. While ‘JEPA’ is a specific architecture (Joint Embedding Predictive Architecture), it is not as ubiquitous as ‘Transformer’ or ‘RNN’. Define it at first occurrence: ‘Joint Embedding Predictive Architecture (JEPA)’.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a logically coherent argument structure, defining the problem (agentic environments), categorizing existing solutions (attributes, domains, synthesis methods), and proposing future directions. The definitions of POMDP, Agent, and Environment in Section 2 are consistent with the usage in later sections (e.g., Section 5 on synthesis and Section 6 on evolution). The taxonomy presented in the Introduction aligns with the detailed sections.

However, there are minor inconsistencies between the textual claims and the summary tables that require clarification to ensure the argument holds together without ambiguity:

1. **Data Source Discrepancy (Section 5.2 vs. Table 2):** The text in Section 5.2 explicitly states that V-JEPA 2 “requires only 62 hours of robot data.” In contrast, Table 2 (Representative methods in neural synthesis) lists the “Data Source” for V-JEPA 2 as “VideoMix22M.” While “VideoMix22M” could theoretically be a dataset name, the “22M” usually implies a massive scale (millions of samples), which conflicts with the “62 hours” claim unless the text specifies that only a 62-hour subset was used. The reader cannot verify the “62 hours” claim from the table alone. This is a minor logical gap where the premise (table data) does not clearly entail the specific quantitative conclusion (62 hours) in the text.
2. **Method Categorization (Section 6.3 vs. Table 4):** The text in Section 6.3 categorizes ToolBench under “Tree Search” and claims it “uses DFS.” Table 4 (Statistics of Trajectory-Centric Offline Evolution Methods) lists ToolBench under “Trajectory Synthesis” with “Augmentation” as the method type. While “Augmentation” can involve tree search, the table does not explicitly confirm the “DFS” mechanism mentioned in the text. The text makes a specific algorithmic claim that is not directly visible in the corresponding table row, creating a slight disconnect between the detailed argument and the summary evidence.
3. **Temporal Logic of Citations:** The Introduction cites “GPT-5.4” and “Gemini-3.1-Pro” as existing models. The bibliography lists these with a publication year of 2025 and access dates in 2026. While acceptable for a preprint, the logical flow of a survey paper typically relies on established, verifiable facts. If the paper is positioning these as the current state-of-the-art benchmarks, the future-dated references (relative to the typical “now” of a survey) should be explicitly framed as “upcoming” or “pre-release” in the main text to avoid the logical implication that these are already widely deployed and evaluated in the same way as older models like WebShop (2022).

These issues are primarily matters of clarity and alignment between text and tables rather than fundamental logical fallacies. The core argument remains sound, but these specific points need tightening to ensure the premises (tables) fully support the conclusions (textual claims).

Action items.

- **[writing]** Section 5.2 claims V-JEPA 2 uses ‘62 hours’ of data, but Table 2 lists ‘VideoMix22M’ as the source. Clarify if the table refers to a subset or if the text needs a specific citation to resolve the apparent scale mismatch.
- **[writing]** Section 6.3 states ToolBench uses DFS, but Table 4 categorizes it under ‘Augmentation’ without mentioning DFS. Align the table’s ‘Trajectory Source’ column or add a note to support the specific DFS claim.
- **[writing]** The Introduction cites GPT-5.4 and Gemini-3.1 as existing models, but bibliography dates are 2025/2026. Explicitly frame these as ‘pre-release’ or ‘forthcoming’ in the text to avoid logical tension regarding the current state-of-the-art baseline.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper’s rhetoric occasionally exceeds the scope of the evidence provided by the survey’s own findings. As a survey, the “evidence” is the state of the literature itself; the overreach occurs when the authors describe the field as more mature, unified, or solved than the literature actually indicates.

First, the abstract and introduction use strong, definitive language (“systematically studies,” “inseparable twin,” “continual evolution”) that implies a robust, established engineering discipline. However, the paper’s own “Takeaway” boxes (specifically 5.1 and 5.3) explicitly admit that key evaluation dimensions like diversity, complexity, and fidelity are “under-researched” and “preliminary.” There is a mismatch between the confident framing of the field as a solved engineering lifecycle and the admission that the evaluation component of that lifecycle is nascent. The language should be hedged to reflect that the field is in an “emerging” or “fragmented” stage rather than a “systematic” one.

Second, the introduction generalizes the mechanism of “continual evolution” to all agentic environments. The text suggests that environments inherently enable this evolution through interaction. Yet, the survey’s own taxonomy in Section 6 reveals that many “Agent Evolution” methods (Memory-Centric, Orchestration-Centric, Trajectory-Centric) rely on static datasets, workflow design, or offline SFT, where the environment is a fixed benchmark rather than a dynamic co-evolving partner. The claim that environments are the “inseparable twin” for *all* evolution pathways overgeneralizes the specific case of Reinforcement Learning (Exploration-Centric) to the entire field.

Finally, the “Future Directions” section presents specific architectural shifts (EaaS, Neural-Symbolic) as necessary evolutions. While these are valid hypotheses, the paper frames them as the logical conclusion of the current state without acknowledging the significant empirical gaps or alternative paths. Given that the survey highlights the lack of standardization and the “preliminary” nature of current evaluation, presenting these specific futures as the definitive path forward is a form of overreach by omission of the uncertainty inherent in the field’s current trajectory.

Action items.

- **[writing]** Abstract claims ‘systematically studies’ and ‘continual evolution,’ but Section 5.3 admits evaluation of diversity/complexity/fidelity is ‘preliminary.’ Replace ‘systematically studies’ with ‘synthesizes emerging research’ and qualify evolution claims to reflect the field’s nascent state.
- **[writing]** Introduction states environments enable ‘continual evolution’ for all agents, but Section 6 shows many methods (Memory/Orchestration-centric) use static data. Narrow the claim to specify this applies primarily to ‘Exploration-Centric’ and ‘Neural-Driven’

paradigms.

- **[writing]** Section 7 presents ‘Environment-as-a-Service’ and ‘Neural-Symbolic’ as inevitable next steps. Frame these as ‘promising research avenues’ rather than ‘critical directions’ to avoid implying a consensus not supported by the current fragmented evidence.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper is a survey of agentic environment engineering, categorizing existing benchmarks, synthesis methods, and evolution paradigms. As a survey, it does not generate new dual-use capabilities, scrape new datasets, or conduct human-subjects research. The paper references existing benchmarks (e.g., WebShop, SWE-Bench, OSWorld) and synthesis techniques (e.g., symbolic code generation, neural world models) but does not provide operational details for exploiting vulnerabilities, generating harmful content, or bypassing safety filters in a way that creates immediate, actionable risk.

The discussion of “de novo synthesis” and “self-play” (Section 5.1, 7.1) describes methodological trends in the field rather than releasing a specific, dangerous tool. The paper explicitly frames these environments as tools for safe, reproducible training and evaluation, often highlighting the reduction of real-world risks (e.g., Section 1). There are no indications of undisclosed conflicts of interest, PII leakage, or license violations in the data sources discussed (which are standard public benchmarks).

While the field of agentic AI carries inherent long-term dual-use potential (e.g., automated hacking or disinformation), this specific paper does not cross the threshold of creating a *foreseeable, non-trivial risk* that is *unacknowledged*. It functions as a taxonomy of the field rather than an operational manual for harm. No specific safety disclosures or mitigations are missing because the paper does not perform the high-risk actions that would necessitate them. The verdict is accept.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

This survey aims to establish a scientific foundation for “agentic environment engineering” by categorizing existing environments, synthesis methods, and evolution paradigms. However, the evidentiary strength of the paper is critically compromised by the inclusion of non-existent or

future-dated models and benchmarks as if they were established facts.

The most severe issue lies in the foundational evidence: the paper cites “GPT-5.4” (2025), “Gemini-3.1-Pro” (2025), “Kimi K2.5” (2026), and “DeepSeek-V3.2” (2025) as current, representative models driving the field. As of the current date, these models do not exist. Citing hallucinated or future-dated artifacts as the basis for a “systematic study” of the current landscape renders the entire taxonomy and comparative analysis suspect. A survey’s primary evidence is the literature it reviews; if the literature cited includes non-existent works, the survey cannot claim to accurately reflect the state of the art. This is not merely a citation error but a fundamental failure of the evidentiary basis for the paper’s central claims.

Furthermore, the paper makes broad claims about the “effectiveness” and “superiority” of various synthesis and evolution methods (e.g., “De Novo Synthesis offers the highest degree of freedom,” “Neural-Driven Evolution represents the environment itself as a learned model”) without providing quantitative evidence to support these assertions. The tables summarizing these methods (e.g., Table 5.1, Table 6.1) list metrics like “Correctness” or “Success Rate” but do not provide the underlying data, variance, or a consistent baseline for comparison. The evidence presented is entirely dependent on the original authors’ single-run reports, which are often not reproducible or comparable across different papers. Without a meta-analysis or a unified evaluation framework, the survey cannot establish that one method is genuinely superior to another; it can only report that different papers claim different things. This lack of rigorous cross-validation means the paper’s conclusions about the “best” approaches are not supported by the evidence presented.

To be accepted, the authors must rigorously verify the existence and publication status of every cited model and benchmark, removing any hallucinated or future-dated entries. Additionally, they must either provide a unified, reproducible evaluation of the cited methods or explicitly qualify their claims as “reported in the literature” rather than established facts, acknowledging the lack of comparative evidence.

Action items.

- **[fatal]** This survey aims to establish a scientific foundation for “agentic environment engineering” by categorizing existing environments, synthesis methods, and evolution paradigms. However, the evidentiary strength of the paper is critically compromised by the inclusion of non-existent or future-dated models and benchmarks as if they were established facts. The most severe issue lies in the foundational evidence: the paper cites “GPT-5.4” (2025), “Gemini-3.1-Pro” (2025), “Kimi K2.5” (2026), and “DeepS

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

This paper is a survey article that synthesizes and categorizes existing research on agentic

environments, benchmarks, and synthesis methods. It does not present new empirical experiments, nor does it compute novel statistical estimates, perform hypothesis testing, or report quantitative results derived from a specific experimental run. Consequently, there are no statistical procedures (e.g., t-tests, ANOVA, regression models, confidence intervals) to evaluate for correctness, assumption violations, or multiple comparison corrections.

The quantitative elements present in the text are primarily descriptive counts (e.g., number of benchmarks, dataset sizes listed in tables) or citations of performance metrics from other works. These are reported as factual attributes of the cited works rather than as results of a statistical analysis performed by the authors. For instance, Table 1 lists dataset sizes (e.g., “500” for WebShop) and binary attributes (e.g., “Partially” observable), which are descriptive facts, not statistical inferences requiring error bars or p-values.

Since the lens of statistical analysis specifically targets the computation, reporting, and interpretation of statistical machinery, and this paper contains no such machinery, there are no statistical errors to flag. The paper correctly avoids making inferential claims where none are warranted by its survey nature.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper presents a comprehensive survey on agentic environment engineering, but the prose occasionally suffers from structural fragmentation and weak signposting that forces the reader to re-parse sentences to recover the main point.

The most significant issue is the inconsistent use of topic sentences, particularly in the synthesis and evolution sections. For instance, in Section 5.1, the discussion on “De Novo Synthesis” begins with a sentence fragment that lacks a clear subject, creating an immediate barrier to entry for the reader. Similarly, in Section 6.3, the paragraph detailing trajectory generation methods launches directly into a list of four approaches without a guiding topic sentence. This forces the reader to infer the paragraph’s purpose only after reading the entire block. A simple rewrite to state the categorization upfront would resolve this friction.

Transitions between sub-sections are also frequently abrupt. In Section 5.2, the shift from “Some approaches” to “Other approaches” within the same paragraph regarding pixel-level modeling feels disjointed, as if two separate thoughts were pasted together. A stronger connective phrase explaining the evolution from limited interaction to long-horizon capabilities would smooth this flow. Additionally, the introduction of specific RL algorithms (PPO, GRPO, DAPO) in Section 3 lacks a bridging sentence that explains their relevance to the “Environment-Agent Alignment” concept being defined, making the sudden technical dive feel unmotivated.

Finally, the abstract fails to fully summarize the paper’s structural contribution. While it lists future directions, it omits the core taxonomy of eight attributes and eight domains

that forms the backbone of the survey. A reader skimming only the abstract would miss the primary organizational framework of the paper. Rewriting the abstract to explicitly mention this taxonomy would ensure it accurately reflects the body’s content.

Addressing these specific structural and transitional issues will significantly improve the readability and flow of the manuscript without altering its scientific content.

Action items.

- **[writing]** The paper presents a comprehensive survey on agentic environment engineering, but the prose occasionally suffers from structural fragmentation and weak signposting that forces the reader to re-parse sentences to recover the main point. The most significant issue is the inconsistent use of topic sentences, particularly in the synthesis and evolution sections. For instance, in Section 5.1, the discussion on “De Novo Synthesis” begins with a sentence fragment that lacks a clear subject, creating an